

Name:

ADM No:

Section:

SEMESTER: MONSOON, SESSION: 2022-2023

Quiz-2

Subject: Basics of Electronics Engineering

Time: 30 Mins

Subject Code: ECI101

Max Marks: 26

Instructions: **Tick only the correct option.**

1. Once in saturation, a further increase in base current will [1 Mark]
(A) cause the collector current to increase
(B) not affect the collector current
(C) cause the collector current to decrease
(D) turn the transistor off
2. When the collector junction in transistors is biased in the reverse direction and the emitter junction in the forward direction, the transistor is said to be in the [1 mark]
(A) active region
(B) cut-off region
(C) saturation region.
(D) none of the
3. The JFET in the circuit shown in **Fig.1** has a maximum saturation current $I_{DSS} = 15 \text{ mA}$ and pinch-off voltage $V_p = -5 \text{ V}$. The value of resistance R_s for the drain current $I_{DS} = 6.4 \text{ mA}$ is? [2 marks]
(A) 156Ω (B) 473Ω (C) 1000Ω (D) 271Ω (E) None
4. If the input to the ideal op-amp of **Fig.2** is a sine wave the output will be [1 Mark]
(A) half wave rectified sine wave (B) A full-wave rectified sine wave
(C) A triangular wave (D) A square wave
5. If BJT is in cut off, V_{CE} is [1 Mark]
(A) 0 V (B) minimum (C) maximum (D) equal to V_{CC} (E) Both (A) and (B)
(F) Both (C) and (D)
6. In a PNP transistor, the majority charge carriers flow- [1 Mark]
(A) Into the transistor at the emitter and base leads
(B) Out of the transistor at the emitter and base leads
(C) Into the transistor at the collector and base leads
(D) Out of the transistor at the collector and base leads
7. The leakage current I_{CBO} flows in- [1 mark]
(A) The emitter, base and collector leads
(B) The emitter and base leads
(C) The emitter and collector leads
(D) The base and collector leads
8. The circuit shown in **Fig. 3** is of the N channel JFET. The pinch-off voltage V_p and the maximum saturation current I_{DSS} are -8 V and 10 mA respectively. What are the values of gate to source voltage V_{GS} and drain current I_D ? [2 Mark]
(A) $+4 \text{ V}$ and 25 mA (B) -4 V and 2.5 mA (C) $+4 \text{ V}$ and 2.5 mA (D) -4 V and 25 mA
9. For the Op-Amp circuit shown in the **Fig. 4**, V_0 is [2 Mark]
(A) -2V (B) -0.5V (C) -1V (D) 0.5V

10. A BJT has a common base forward current gain of 0.98. Its common emitter forward current gain will be [1 mark]
 (A) 51 (B) 99 (C) 100 (D) 49
11. Which of the following statement is true about FET? [1 Mark]
 (A) It has high output impedance (B) It has high input impedance
 (C) It has low input impedance (D) It does not offer any resistance
12. To use FET as a voltage-controlled resistor, in which region it should operate? [1 Mark]
 (A) Ohmic region (B) Cut off (C) Saturation (D) Both cut off and saturation
13. An ideal op-amp is an ideal [1 Mark]
 (A) voltage controlled current source (B) voltage controlled voltage source
 (C) current controlled current source (D) current controlled voltage source
14. In the Fig. 5, determine I_E and I_C , if $\beta = 49$. [2 Mark]
 (A) 1.0 mA , 0.98 mA (B) 10.0 mA , 9.8 mA (C) 1.3 mA , 1.27 mA (D) 9.49 mA , 9.3 mA
15. If the op-amp in Fig. 6 is ideal then V_0 is [2 Mark]
 (A) Zero (B) $(V_1 - V_2) \sin wt$ (C) $-(V_1 + V_2) \sin wt$ (D) $(V_1 + V_2) \sin wt$
16. A transistor connected in common-base configuration has- [2 Mark]
 (A) a high input resistance and a low output resistance
 (B) a low input resistance and high output resistance
 (C) a low input resistance and a low output resistance
 (D) a high input resistance and a high output resistance
17. Find out I_E and V_C for the given circuit shown in Fig. 7. [2 Mark]
 (A) 1.86 mA, 3.3 V (B) 13.2 mA, 16 V (C) 2.0 mA, 14.7 V (D) 2.8 mA, 13.3 V
18. Calculate the values of all three currents (in mA) in the circuit shown in Fig. 8. Neglect V_{BE} [2 Mark]
 (A) $I_B = 0.007$, $I_C = 0.7$, $I_E = 0.707$ (B) $I_B = 0.006$, $I_C = 0.6$, $I_E = 0.606$ (C) $I_B = 0.008$, $I_C = 0.8$, $I_E = 0.808$
 (D) $I_B = 0.009$, $I_C = 0.9$, $I_E = 0.909$

